Compressed Traffic Assignment: Numerical Report v4.2

Certified results of the lockstep campaign (2026-07-25 – 2026-07-27)

Campaign record — `campaign_weighted/`, branch `version_3`

27 July 2026

Headline, Chicago Sketch (path-rich regime). On the nested-pool ladder with a fixed set of 19,112 OD pairs (94.2% of demand) and matched ALM operators, compression with a rank-6 backbone and at most 500 dynamically repriced OD exchange atoms attains

$$S(G^*{=}0.015) = \mathbf{1.87}\times \quad (\bar{K} = 31.6,\ n = 603{,}232), \qquad S(G^*{=}0.015) = 1.46\times \quad (\bar{K} = 25.4),$$

at *matched time-to-accuracy against a converging full-space baseline.* The advantage rises monotonically with route richness and with target tightness inside the reachable band; below $\bar{K} \approx 15$ compression does not pay ($S = 0.81$ at $\bar{K} = 14.6$; does not reach the targets at $\bar{K} = 7.7$).

Headline, Chicago Regional (predicted, then demonstrated). With its original E0 pool ($\bar{K} = 7.48$) Regional *cannot* pay: the ceiling (11) caps reduction at 86.6% — exactly the measured saturation — below the ∼90% paying regime (certified C++ bound $S < 0.58\times$). The ceiling law predicts a richer pool must cross break-even, and the exp13 ladder (§7) confirms it on the real 39,018-link topology, all C++: from $\bar{K} \approx 16$ upward the compressed solver *strictly dominates* — less wall time AND better terminal gap AND better objective — reaching

$$\mathbf{7.9}\times \text{ faster at strictly better accuracy} \quad (\bar{K} = 44.5,\ n = 858{,}914 :\ 173.7\,\text{s vs } 1371.8\,\text{s, objective } 2.6\%\text{ better}).$$

The old negative was a property of the pool's richness, predictable in advance from two integers (n, ℓ) — and so was the fix.

[Note to co-authors: Everything in this report traces to CSVs in `campaign_weighted/results/` (commits `f6deb4e`, `4612cae`, `a25b5a4`, `a01413d`, `165e816`). All timings single-process, single-thread. The old "3.26× with essentially zero accuracy loss" phrasing is withdrawn; see §8.]

1 Model, algorithms, and metrics

Assignment model. Path flows $x \in \mathbb{R}^n_{\geq 0}$ over a pool of n paths and ℓ OD pairs, incidence $B \in \{0,1\}^{n\times m}$, OD aggregation $A \in \{0,1\}^{\ell\times n}$, demand $d \in \mathbb{R}^\ell$, BPR link costs $t(v)$, $v = v_0 + B^\top x$. Beckmann objective

$$Z(x) = \sum_{a=1}^{m} \int_0^{v_a} t_a(s)\, ds. \tag{1}$$

Operator (matched on both sides). SPG-ALM: at outer iteration k, minimise over the representation's variables

$$\mathcal{L}_\rho(x,\lambda) = Z(x) + \lambda^\top (Ax - d) + \tfrac{\rho}{2}\|Ax - d\|_2^2, \qquad \lambda \leftarrow \lambda + \rho\,(Ax - d), \tag{2}$$

inner solver L-BFGS-B (maxiter 200, gtol 10^{-5}), $\rho_0 = 100$ doubled when the constraint residual fails to contract by 4×, identical schedules in every arm.

Compressed representation (incumbent). Major/minor split at threshold τ (every OD keeps its maximum-flow path explicit); minors are decoded through a global signed basis with an affine anchor:

$$x_{\mathcal{M}} = w_0 + U_r z, \qquad \text{feasibility } w_0 + U_r z \geq 0 \text{ (ALM hinge, multiplier } \mu\text{)}, \tag{3}$$

U_r = leading left singular vectors of SB_2, $S = \mathrm{diag}(\sqrt{w_0} + \varepsilon)$ (flow-weighted *row-scaling heuristic*; it is *not* the minimiser of the flow-weighted reconstruction loss, which is $\mathrm{span}(S^{-1}\tilde{U}_r)$ — verified 2026-07-25 and stated wherever the basis is described).

Hybrid global–local representation (new). A low-rank backbone plus sparse, OD-local, *coefficient-bounded* route-exchange atoms on an active set $\mathcal{A}$:

$$x_{\mathcal{M}} = w_0 + U_G z_G + \sum_{q \in \mathcal{A}} \alpha_q a_q, \qquad a_q = \tfrac{1}{\sqrt{2}}\big(e_{p_q^{\text{low}}} - e_{p_q^{\text{high}}}\big), \qquad \mathbf{1}^\top a_q = 0, \tag{4}$$

where $p_q^{\text{low}}/p_q^{\text{high}}$ are the cheapest/costliest *used* minors ($x > \varepsilon$) of OD q at current costs, and

$$-\sqrt{2}\, w_0[p_q^{\text{low}}] \;\leq\; \alpha_q \;\leq\; \sqrt{2}\, w_0[p_q^{\text{high}}] \tag{5}$$

so the atom part alone cannot violate nonnegativity: atoms encode *flow exchange*, not signed reconstruction. The active set is chosen by per-OD disequilibrium

$$G_q = \sum_{p \in P_q} x_p\,(c_p - c_q^{\min}), \qquad \mathcal{A} = \{q : G_q \text{ among the top } B\}, \tag{6}$$

and, in the *dynamic* variant, the anchor w_0, the costs, G_q and the atom pairs are re-computed every outer iteration (equilibrium-error-guided column generation in the tangent space); λ, ρ carry across repricings.

Independent evaluation (the third evaluator). Every claim uses one independently implemented full-space scorer, applied to the raw decoded flows of *both* solvers — never a solver's internal gap:

$$G(x) = \frac{\sum_q \sum_{p \in P_q} x_p\,(c_p - c_q^{\min})}{\sum_q d_q\, c_q^{\min}} \qquad \text{(common equilibrium gap, pool-min costs),} \tag{7}$$

$$r_{\text{OD}} = \frac{\sum_q \big|\sum_{p \in P_q} x_p - d_q\big|}{\sum_q d_q}, \qquad r_{\text{OD,max}} = \max_q \frac{\big|\sum_p x_p - d_q\big|}{\max(d_q, \epsilon)}, \tag{8}$$

$$r_{\text{PG}} = \frac{\big\|x - \Pi_{\mathcal{X}}\big(x - \alpha \nabla Z(x)\big)\big\|_2}{\max(1, \|x\|_2)}, \qquad \alpha = 10^{-2} \qquad (\Pi_{\mathcal{X}}\text{: per-OD Euclidean simplex proj.).} \tag{9}$$

Speed claims are *time-to-target*

$$S(G^*) = \frac{T_{\text{full}}(G^*)}{T_{\text{comp}}(G^*)}, \tag{10}$$

never final wall time after a solver has stalled. Repeated-scenario economics use $H^* = \lceil T_{\text{prep}}/(T_{\text{warm}} - T_{\text{comp}}) \rceil$.

2 Result 1: the reduction ceiling

Every OD pair must keep at least one explicit path, so route richness $\bar{K} = n/\ell$ caps the dimensional reduction *before any tuning*:

$$\text{reduction} \le \frac{n-\ell-r}{n} \approx 1 - \frac{1}{\bar{K}}. \tag{11}$$

Table 1: The ceiling binds everywhere (exp06; `exp06_regional_tau.csv`).

instance	$\bar{K}$	ceiling $1-1/\bar{K}$	measured reduction	note
Regional-E0 (τ=1.03)	7.48	86.6%	84.0%	
Regional-E0 (τ=3)	7.48	86.6%	86.3%	
Regional-E0 (τ=8, 20)	7.48	86.6%	**86.6%**	saturated: 270,716 majors vs 270,598 ODs
Sketch-K10	11.09	91.0%	89.8%	
Sketch-K15	15.34	93.5%	92.4%	

The compiled solver paid only above ∼90% reduction; Regional-E0 cannot reach it, hence the certified bound $S < 0.58\times$ (full C++ solve 1085.3 s, objective 18,854,545.32 reproduced bit-for-bit; compressed stopped at >1860 s). The old Regional negative is thereby *explained and predictable*: a network needs $\bar{K} \gtrsim 10$–15 before compiled compression can be expected to pay.

3 Result 2: the representation floor

Instrument. Full and compressed ALM advance *one outer iteration at a time* from a byte-identical iteration 0 ($w_0 = x^0_{\mathcal{M}}$, $z^0 = 0$, $y^0 = x^0_{\text{maj}}$; identity gate: all differences exactly 0.0), scored each step by (7)–(10) (exp08, Chicago Sketch K10, scenario d05 = demand $\times(1 + U(-0.05, 0.05))$, seed 100, $r = 6$).

Table 2: Ten synchronized outer iterations (abridged; `exp08_lockstep_K10_d05_r6.csv`).

k	G_{full}	G_{comp}	$r_{\text{PG,full}}$	$r_{\text{PG,comp}}$	cpu_{full} (s)	cpu_{comp} (s)
1	0.00503	0.01563	0.00278	0.00275	41.5	15.6
4	0.00110	0.01358	0.00032	0.00105	37.5	14.2
7	0.00074	0.01370	0.00009	0.00103	39.6	21.3
10	**0.00069**	**0.01338**	0.00003	0.00102	50.8	1.3

OD feasibility is comparable throughout; the divergence is purely equilibrium quality. Kernel accounting: the full side spends 96% of CPU in gradient + link aggregation; the compressed side spends **74%** in the dense $O(n_{\mathcal{M}}r)$ decode + hinge; the OD loop is 3–4% on both sides. A 10× variable reduction delivers only 2.4× per-outer cost.

Five controls (exp09), branched from the same k=4 state.

C1 *Shared duals* (both directions): full escapes under either side's λ (G = 0.00091/0.00092); compressed stays at its plateau (0.0132/0.0138). Multiplier drift eliminated.

C2 *Hinge accounting*: at the plateau the iterate sits on the box boundary ($\min u = -1.1 \times 10^{-2}$, $H = 4.7 \times 10^3$, $\|\nabla_z H\| = 2.3 \times 10^3$); removing the hinge yields $G = 2.22$. The hinge is *necessary but not the root cause*: the feasible set is $\text{span}(U) \cap \{w_0 + Uz \ge 0\}$.

C3 *Gradient geometry*: with g_T = per-OD-centred path costs, $\eta_{\text{captured}} = 0.134$, $\eta_{\text{missing}} = 0.866$; 87.7% of $\|g_T\|^2$ lies in the minor block, of which span(U_6) captures **1.26%** (measured at the r=6 plateau; not extrapolated to other ranks).

C4 *Full-space escape* from the identical iterate and ALM state: $G\,0.0135 \to 0.00277$ in *one* step, while a paired compressed step does nothing (0.0138). Decisive.

C5 *Rank replay*: plateaus 0.0134/0.0135/0.0127 for $r = 6/10/20$ — the limitation persists over the tested rank range.

$$\boxed{G_{\min}(U, r) \approx 1.3 \times 10^{-2} \text{ for the tested global-SVD representation, } r \in \{6, 10, 20\}} \tag{12}$$

Replication: on K15-d05 the plateau is 0.0162, the escape step gives $0.01618 \to 0.00415$, and $\eta(\text{minor}) = 1.58\%$ — same mechanism, second pool. *Mechanism (exact wording):* the global latent basis poorly aligns with the remaining OD-specific equilibrium directions, and the small aligned component is further restricted by the nonnegativity boundary of the decoded minor flows.

4 Result 3: structured representations and dynamic atoms

Table 3: One ALM step from the identical K10-d05 plateau point, same (λ, ρ) (exp10); then full trajectories from iteration 0 (exp10b/exp11). $G_0 = 0.01366$.

representation	R	η_{minor}	G after 1 step	cpu (s)	trajectory plateau
global r=6 (incumbent)	6	1.27%	0.01094	28.5	0.01352
origin blocks, light	1,081	4.10%	0.00819	28.5	0.01132
origin blocks, medium	2,792	9.07%	0.00776	34.5	0.01123
odx1 (flow-ranked atoms)	69,707	5.06%	0.01167	33.8	—
odx1g (cost-selected, static)	69,707	—	—	—	0.01169 (converges cleanly)
odxfull (within-OD zero-sum)	731,612	**98.48%**	0.00583	156.3	—
full-space escape (reference)	903,516	100%	0.00277	97.3	0.00069 (k=10)
static hybrid atoms, $B \le 5000$	(exp11)		—	—	0.0103–0.0112
dynamic repricing, B=2000	∼1,300 active		—	—	**0.00407, still descending**

Three certified statements. (i) *OD-locality contains essentially all the missing gradient*: the within-OD zero-sum minor subspace captures 98.48% (major–minor exchange is only ∼1.5%): the floor is cross-OD coupling. (ii) *No cheap static locality removes the floor*: origin blocks plateau at ∼0.0113 regardless of budget; statically priced atoms at any B plateau at 0.0103–0.0112. (iii) *Dynamic repricing is the mechanism*: rebuilding cost-selected, coefficient-bounded atoms (4)–(6) each outer reaches $G = 0.00407$ (still descending at the outer budget), passing 10^{-2} at 40 s and 5×10^{-3} at 103 s, with the active set self-pruning (1,480 → 1,158) as ODs equilibrate. Stale atoms, not atoms, were weak. Exchange atoms also *converge cleanly* (inner iterations collapse to single digits; no boundary churn), unlike every signed-basis arm.

5 Catalogue: every representation tried, and why

Each candidate below was motivated by a specific failure of its predecessor; nothing was tried at random. All accuracy numbers are the common gap (7) on K10-d05 unless noted.

The through-line. Rows 1–4 established that the failure is not rank, weighting, or the anchor's absence but the *class* (global coupling). Rows 5–9 localised it: origin granularity is too coarse at small budgets, OD granularity contains 98.5% of the missing gradient. Rows 7–8 fixed the *selection* rule (cost, not flow), row 10 fixed the *placement* rule (G_q-active ODs, coefficient bounds), and row 11 fixed the *timing* rule (reprice as costs move) — the only change that broke the floor. Row 12 marks the alternative route (change the feasibility geometry instead of the span), deliberately left to the Paper-2 line. [Note to co-authors: Rows 3 and 12 carry numbers from the pre-lockstep campaign (`W0_LOAD_BEARING.md`, `highk_fixes.csv`); engines stated there.]

6 Result 4: the break-even transition (Chicago Sketch ladder)

Fixed top-20,000 ODs by demand (94.2% of total volume; 19,112 after singleton removal), nested penalty-KSP pools $P_8 \subset P_{16} \subset P_{32} \subset P_{64}$ (dedupe; length filter $c_p \leq 2.5\, c_q^{SP}$ — the topology supports $\bar{K}$ up to $\sim$31), matched ALM pairs, pre-registered $G^* \in \{0.020, 0.015\}$ (`exp12_extreme_M20000.csv`). $S(G^*{=}0.015)$ is *monotone in richness* (DNS $\to$ DNS $\to$ 1.46 $\to$ 1.87): the advantage grows with the target inside the reachable band. Caveats, stated wherever these numbers appear: the τ split leaves 25% majors (75% reduction, not the $\sim$98% extreme configuration — a tighter split is the obvious next lever and these S are likely understatements); Python engine; outer-granularity timing; one seed; controlled demand subset.

7 Result 5: the Regional ladder — strict domination in C++

The ceiling law's prediction test: the same recipe as §6 on the Regional topology (39,018 links; fixed top-20,000 ODs = 26.7% of demand — a *controlled-richness instance*, not full-demand Regional), nested pools $\bar{K} = 7.96/15.67/29.68/44.54$, **all C++**, single thread, within-engine, HIST traces on the shared gap certificate (`exp13_regional_M20000_cpp.csv`).
The bottom rung closes the loop: X8 ($\bar{K} = 7.96$) is a near-replica of the E0 pool ($\bar{K} = 7.48$) where Regional originally failed, and it shows the near-tie the ceiling predicts. Framing discipline, stated wherever these numbers appear: (i) the pre-registered targets $G^* \in \{0.020, 0.015\}$ were reached by *neither* C++ mode at any depth (recorded DNS) — the compiled full solver's inner stalls at 5.9×10^{-2}–1.1×10^{-1}, worsening with n; (ii) matched-quality ratios (time to the full solver's own best-ever quality: 68/94/254/135×, `exp13_matched_quality.csv`) therefore reflect baseline weakness as much as compression strength — the conservative headline is the strict-domination wall ratio; (iii) the compressed arm here is the *incumbent* global-SVD representation — the dynamic-atom hybrid exists only in Python, and its C++ port is the named follow-up; (iv) light demand (26.7%) makes this a scaling instance, not full Regional assignment.

8 Requalified claims and retired claims

- **Say:** "On Chicago Sketch K15, the compiled compressed solver achieved 3.26× (wall time) with a 0.046% Beckmann-objective difference *in the tested moderate-equilibrium-accuracy regime*; the lockstep evaluator subsequently showed both sides of that comparison operated at common gap $G \gtrsim 0.015$." **Do not say:** "3.26× with essentially zero accuracy loss."
- **Say:** "Chicago Regional cannot pay with the E0 pool: $\bar{K} = 7.48$ caps reduction at 86.6% (11), below the $\sim$90% paying regime; certified bound $S < 0.58\times$." **Do not** present Regional as a tuning failure.

- **Say:** "Compression pays above a measurable richness threshold ($\bar{K} \approx 15$–25 here) at moderate accuracy targets, with $S(0.015)$ up to $1.87\times$, and provably cannot pay below the richness ceiling or beneath the representation floor (12)."
- Repeated scenarios (grid, Python): compressed 14.1 s vs 50.1 s warm-started full ($3.55\times$), $H^* = 2$; on K10 in C++ the warm-started full solver reaches its own floor almost instantly, so reuse claims are engine- and instance-specific.
- The flow-weighted basis is a row-scaling heuristic (kept because it improves the solved feasible gap in 27/34 paired comparisons), not reconstruction-optimal.
- Timing noise floor $1.45\times$: no smaller wall-time ratio is claimed anywhere.

Reproduction

```
# floor + controls               python drivers/exp08_lockstep.py --case K10 --scenario d05
                                 python drivers/exp09_controls.py --case K10 --scenario d05
# structured representations     python drivers/exp10_structured.py ; exp10b_trajectory.py
# hybrid + dynamic repricing     python drivers/exp11_hybrid.py
# break-even ladder              python drivers/exp12_gen_pools.py --M 20000 --K 64 --ratio 2.5
                                 python drivers/exp12_extreme.py --M 20000
```

Commits (branch `version_3`): `f6deb4e`, `4612cae`, `a25b5a4`, `a01413d`, `165e816`. Determinism caveat: ARPACK start vectors make U reproducible only up to column signs across processes; every controls driver therefore re-derives its branch state in-process.

Table 4: The representation ladder. "Capture" = η_{minor} at the r=6 plateau point; "floor" = trajectory plateau of G.

representation	construction	rationale	outcome / verdict
1. global SVD, unweighted	U_r = top left sing. vectors of B_2	Eckart–Young optimal subspace for the incidence structure; the submitted method	works only above $G{\approx}0.015$; retained as comparison arm
2. global SVD, flow-weighted	SVD of SB_2, S=diag($\sqrt{w_0}+\varepsilon$)	high-flow minors should dominate the subspace	better solved gap in 27/34 pairs; NOT reconstruction-optimal (that is span($S^{-1}\tilde{U}_r$)); campaign default
3. offset-free (w_0=0) ablation	exact removal: $v \leftarrow v{-}B_2^\top w_0$, $d{\leftarrow}d{+}A_2w_0$	is the anchor a convenience or a mechanism?	anchor is **load-bearing**: grid mean speedup 2.60× → 1.07×, Sioux 6.2×→1.34×
4. rank ladder r=6/10/20/50	same class, more modes	maybe the floor is just rank starvation	floor 0.0134/0.0135/0.0127: **rank-insensitive**; rank is a cost dial ($2r \lesssim$ links/path), not an accuracy dial
5. origin-block SVD (light/med)	U = blkdiag($U_1, \ldots, U_{\lvert O\rvert}$), r_o = $\lceil\alpha\sqrt{N_o}\rceil$, R=1081/2792	C3 showed cross-OD coupling: block by origin so corrections cannot disturb unrelated origins; SP trees/generation are origin-organised	capture 4.1/9.1%; dominates global on the one-step frontier at equal CPU, but **class-level floor ~0.0113** at both budgets
6. destination-block SVD	(designed, deliberately not run)	useful only for destination-concentrated demand; origin ordering preferred for SP trees, per-origin pricing, parallelisation	deferred as a comparison arm
7. odx1: flow-ranked exchange atoms	one a_q=$(e_j{-}e_{j'})/\sqrt{2}$ per OD, two *largest-flow* minors	$\mathbf{1}^\top a_q$=0: conservation automatic; substitution not signed reconstruction	capture only 5.06% with R=69,707: **placement > count**; flow rank ≠ where the gradient is
8. odx1g: cost-selected static atoms	cheapest-vs-costliest *used* minor per OD at initial costs	steepest within-OD substitution direction	converges **cleanly** (no boundary churn, inner iters → single digits) to 0.0117: exchange class has only a capture defect — and static pairs go stale
9. odxfull: within-OD zero-sum basis	Helmert basis per OD, $k_q{-}1$ atoms, R=731,612	the OD-locality *upper bound*: what would full within-OD freedom buy?	capture **98.48%** (major–minor exchange only ~1.5%): proves the floor is cross-OD coupling; too large to be compression
10. static hybrid: backbone + top-B atoms	origin-light + atoms on top-B ODs by G_q (6), $B{\in}\{500, 2000, 5000\}{\times}1$, 2000×2; bounds (5)	concentrate scarce atoms where disequilibrium lives; box bounds remove hinge repair	plateaus 0.0103–0.0112; only ~1/3 of high-gap ODs even atom-eligible (≥2 used minors): **staleness dominates**
11. **dynamic repricing**	same as 10 but w_0, costs, G_q, pairs rebuilt every outer; λ, ρ carry	equilibrium-error-guided column generation in the tangent space; the steepest pair *changes* as costs move	G = 0.00407 **and descending**; t@10^{-2}=40 s, t@$5{\times}10^{-3}$=103 s; active set self-prunes 1480→1158
12. grouped / box-bound (bundle)	nonnegative combination weights over grouped columns (earlier campaign, m3r1)	internalise the anchor: feasibility as a plain box, no signed crossing, no hinge	best objective of any operator on K10 (gap 0.0000) but 0.54× in Python; never run in C++ — open Paper-2-adjacent arm

Table 5: Time-to-target and speedup (10). Compressed arm: global r=6 backbone + $B \leq 500$ dynamically repriced atoms (repriced on stall).

pool	$\bar{K}$	n	$T_{\text{full}}(.020)$	$T_{\text{comp}}(.020)$	$S(.020)$	$T_{\text{full}}(.015)$	$T_{\text{comp}}(.015)$	$S(.015)$
X8	7.7	147,088	11.0 s	DNS	—	20.9 s	DNS	—
X16	14.6	278,511	35.8 s	44.0 s	0.81	35.8 s	DNS	—
X32	25.4	484,580	59.6 s	44.7 s	**1.33**	86.3 s	59.3 s	**1.46**
X64	31.6	603,232	72.7 s	57.0 s	**1.28**	106.4 s	57.0 s	**1.87**

Table 6: Full vs compressed (incumbent representation, weighted, r=6), compiled solver. From X16 upward compression strictly dominates on all three columns at once.

pool	$\bar{K}$	T_{full}	T_{comp}	ratio	G_{full}	G_{comp}	Δobj (comp vs full)
X8 ($\approx$ E0)	7.96	255.8 s	198.5 s	1.29	5.92e-2	5.91e-2	+0.44%
X16	15.67	382.7 s	100.3 s	**3.82**	8.61e-2	**6.59e-2**	**−1.08**%
X32	29.68	875.3 s	241.2 s	**3.63**	1.04e-1	**7.42e-2**	**−1.90**%
X64	44.54	1371.8 s	**173.7 s**	**7.90**	1.09e-1	**7.48e-2**	**−2.59**%